

DIVYANSH KHARE

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EXPERIENCE

MADE

Jun 2026 – Present

Co-Founder & Founding Engineer

San Jose, CA

- Designed a demand-validated production model that replaces forecast-based manufacturing with pre-order-triggered batches: cut demand-to-delivery from an industry-standard 3-5 months to 4-8 weeks, with zero unsold inventory by construction.
- Architected an intermediate-representation translation layer converting AI-generated garment designs directly into production-ready DXF pattern files and techpacks, eliminating manual handoff to manufacturing partners across 12 garment silhouettes, 10 fabrics, and 8 fabrication techniques.
- Architected an MCP integration layer exposing the platform's catalog, configurator, and order management to AI assistants like Claude and ChatGPT under the same manufacturability and authorization guardrails as the web app — a zero-install storefront and back office for every assistant.
- Made the founding technical call to de-risk core production accuracy before demand infrastructure: scoped the pilot around a manual Stripe checkout instead of the automated threshold trigger, shortening time-to-first-verified-garment while preserving a clean upgrade path to the full model.

Teaching Lab Studio

Mar 1, 2026 – May 29, 2026

Software Engineer III • Tech Lead, SCM

Remote

Departed when the company made the strategic decision to offshore its engineering team, shortly after the SCM pilot went live.

- Sole architect and tech lead of SCM: took it 0→1 to a live production pilot with a large customer while owning engineering, product strategy, and forward-deployed customer work.
- Designed and built A2UI, a declarative agent-to-UI protocol letting connected agents render live React components from a bounded LangGraph activity library and the user's capability manifest — architected so every emission resolves through the per-invocation grant model, keeping agents unable to widen the FERPA surface.
- Re-architected the FERPA boundary from in-flight scrubbing to structural isolation: the model sees only opaque per-run references and numeric metrics, while student PII rehydrates in the authorized user's browser behind a per-request check, so names never reach the AI region.
- Built the capability registry, skill launcher, and per-invocation authorization model end-to-end (NestJS, Prisma, Postgres, Next.js, React): time-bound grants scope to a staff/capability/classroom/material tuple, resolve server-side, and back 7 FERPA-mapped audit events.
- Built SCM into an identity provider for connected AI agents on Python/FastAPI + LangGraph: single-use grants read scope from the database row, never the agent's request, and a per-failure-class retry policy caps each workflow at ≤2 LLM calls.
- Engineered the Temporal runtime with human-in-the-loop approval gates: deterministic, replayable workflows carry a re-validation closure, so an authorization that expires mid-review can never write to the database once approval lands hours later.

Union Pacific Railroad

Mar 2023 – Feb 2026

Software Development Engineer II • Mechanical Systems

Omaha, NE

- Led the freezer-car modernization that reversed an 89% revenue-loss trajectory into \$8M+/year at a 45% efficiency gain: built a deterministic Kafka correlation engine joining wayside, AEI, and CAD signals on trainID, closing sensor-coverage gaps (AEI-mismatch humped-car detection recovered via Yard Systems APIs) and applying an ML-derived health threshold to flag derailment-risk bearings for crew intervention.
- Designed the operator-facing consoles for Automatic Equipment Identification and Wayside Detection, replacing raw log-scraping with consistency-aware React UIs and live status surfacing — yard staff read equipment health at a glance, and data accuracy rose 85%.
- Owned the CADX web applications end to end, the interface train crews read compliance-critical directives through, from track-blockage bulletins to severe-weather authorities, where a misread screen is a safety event; also owned those features in the 2M-line legacy C++ backend, no AI assistant.
- Fixed CADX's auth architecture after it exposed SSO session cookies to the browser: routed the frontend through a server-side proxy holding the session, moved to httpOnly cookies, and added CSRF double-submit protection so tokens never reach client JavaScript.
- Shipped field inspection and reporting tooling as an offline-first, touch-first hybrid app on ruggedized Zebra handhelds, built for gloved hands, glare, and no signal; edge caching and async sync cut inspection and reporting time 60%.
- Built a management-scoped MCP-based AI assistant over the reconciled dataset, collapsing a day-to-week analyst SQL lookup across three segregated databases into a real-time answer — field staff now locate a humped freezer car fast enough to save cargo that previously spoiled unlocated.
- Decomposed a legacy C++ monolith into modular services, a 70% gain in real-time responsiveness across three applications, and stood up structured telemetry at >90% coverage, cutting downtime 50% and deploys 40% faster.

STARTUPS

2022 – 2024

Passion projects that started as personal exploration and grew into real ventures with genuine traction.

AetherisVis

Co-founder & CTO

San Diego, CA

Founded AetherisVis and sold it to a commercial gym brand within the same month, delivering the flagship product as sole full-stack engineer.

- Shipped a consumer-facing 3D product configurator for a luxury gym brand as sole full-stack engineer (live at spxgymdesign.com): buyers customize steel finishes, upholstery, and stitch patterns with instant 4K/8K feedback and smooth multi-angle zoom/pan, translated from a UI/UX designer's pixel-perfect comps into production React/Next.js.
- Delivered the configurator as a one-week commercial build embedded in SPX's sales motion, not a demo: client teams present it with no engineer in the room, configurations save and reload, and it degrades gracefully on a hotel-lobby iPad.
- Built a data-first component framework (35+ reusable React/TypeScript components, NextUI) covering every configurator's shared anatomy — scene management, camera controls, material selectors — turning the SPX build into an assembly problem, not an invention one, and cutting UI prototyping time 60% across client projects.
- Engineered a frame-budget scheduler for the renderer: material swaps preview instantly at proxy resolution while full 4K/8K textures stream in only on idle frames within a fraction of the ~16ms budget, holding 60fps under concurrent camera drags and material changes.
- Compressed 8K TwinMotion renders to WebP for 70% faster loads with stitching detail intact at 150% zoom, then added lazy loading, hardware-accelerated pan/zoom, and ARIA/focus management for a touch-friendly, accessible experience.

VortexeAI

Co-founder & CEO

San Jose, CA

Reached 200+ daily active users and secured a TechStars invitation within four months of launch — ultimately opted not to join, staying focused on full-time engineering work.

- Took VortexeAI 0→1 in four months, building the product surface: a low-code React/Next.js node-graph editor where non-engineers compose AI workflows by dragging, connecting, and configuring nodes, with live validation and inspector panels that keep a complex execution model legible.
- Architected the orchestration engine as a directed graph rather than letting an LLM plan its own execution: topological sort and dependency resolution schedule independent branches concurrently and isolate exactly which steps invalidate on failure, making reruns reproducible at 99.9% execution reliability atop non-deterministic model calls.
- Built a dependency-aware memory router so multi-step LLM chains receive only the state each node declared it needed, avoiding the cost and latency of stuffing full history into every call while keeping failures traceable to a single graph edge.
- Drove product-market fit through 100+ customer interviews and five pivots over seven months with a three-person team, converging on a low-code abstraction that cut onboarding time 40% and scaled to 50+ deployed automation patterns.

Overwhelming Studios

Founder & CTO

Campbell, CA

- Shipped two Unreal Engine 5 mobile titles with cloud auth and cross-device sync for 1,500+ players (30% faster loads, 25% better retention), earning direct recognition from Ant Rogers (Head of UI/UX, PlayStation) and reaching PlayStation Studios' promotion-support stage before visa constraints halted progression.

PROJECTS

Closed Circle

[closed-circle.app](#)

Solo Developer

Next.js 15, React 19, TypeScript, Tailwind v4, Supabase/Postgres

- Designed and shipped an invite-only accountability network for founders end to end as sole engineer and designer, from product definition through live production deployment: built the entire design language as one authoritative stylesheet, 77 tokens and 280 component classes across 7 routes, dark-only, on Next.js 15 and Tailwind v4 with hand-vendored Radix primitives.
- Wrote the feed ranker as a product argument, not an engagement optimizer: scored items on obligation and time-decayed silence so an unanswered post rises and a settled one sinks, with engagement log-damped and hard-capped beneath every other signal, then surfaced that reasoning in the UI as per-item chips so users can read why anything is where it is.
- Pushed every business invariant into Postgres instead of React across 10 migrations, 21 functions, 10 triggers, and 30 RLS policies: an advisory lock holding a hard 500-seat cap under concurrent signups, partial unique indexes enforcing one invite per member, and a composite primary key that makes reaction farming impossible from any client.

EDUCATION

University of California, Santa Cruz, B.S. Computer Science: Computer Game Design

Aug 2022

SKILLS

Frontend / UI: React, Next.js, TypeScript, Tailwind, Radix/shadcn, NextUI, design systems, component architecture, accessibility (ARIA/WCAG), responsive & touch-first UI, 3D configurators, generative & agent-emitted UI (A2UI)

Backend & Languages: Node.js, NestJS, FastAPI, Spring Boot, Python, Java, C++, C, C#

AI / Agentic: LangGraph, Anthropic API (Claude), MCP, Temporal workflows, multi-agent orchestration, eval pipelines

Infrastructure: PostgreSQL, Prisma, Temporal, Docker, Kubernetes, AWS, Elasticsearch, Redis, Supabase